

Device Provisioning & Firmware

Formal Technology Names & Architectural Definitions

Version 1.0 — Proposed terminology for the standalone device-engineering software

Purpose: This document establishes a formal vocabulary for the standalone software that configures supported microcontroller boards, defines sensors and parameters, assigns and validates pins, generates firmware, compiles and provisions firmware, verifies the resulting device, and provides a controlled future integration boundary for gateways.

Terminology Status

The constructs in this document are proposed formal names for the new standalone software. They are intentionally separate from the formal Maintain AI constructs defined in “Maintain AI — Formal Technology Names & Architectural Definitions v1.1.” The terminology follows the same architectural style but does not redefine or replace Maintain AI terminology.

1. Formal Name Reference Chart

No.	Acronym	Formal name	Short definition
1	DPFIA	Device Provisioning & Firmware Integration Architecture	Umbrella architecture connecting hardware definition, sensor configuration, firmware generation, provisioning, verification and future gateway integration.
2	HDCRM	Heterogeneous Device Capability Representation Model	Structured representation of supported boards, processors, pins, interfaces, protocols and hardware capabilities.
3	SIRM	Sensor Interface Representation Model	Structured representation of sensors, measurements, interfaces, wiring requirements, libraries and electrical constraints.
4	DCRA	Device Configuration Representation Architecture	Canonical representation of a complete device configuration, including board, machine/device identity, sensors, parameters, pins, sampling and communication settings.
5	APVA	Automated Pin Validation & Assignment Architecture	Automatically assigns compatible pins and detects conflicts, reserved pins and interface incompatibilities.
6	HWVA	Hardware Wiring Validation Architecture	Produces deterministic wiring requirements and validates configured connections against board and sensor definitions.
7	FGA	Firmware Generation Architecture	Transforms validated device configuration into board-specific, sensor-aware firmware source.
8	FCPA	Firmware Compilation & Provisioning Architecture	Compiles generated firmware and provisions it onto the connected physical board using supported toolchains.
9	DVA	Device Verification Architecture	Verifies that a provisioned device boots, communicates and produces expected sensor telemetry.
10	LTA	Local Telemetry Abstraction Architecture	Normalizes locally observed device readings into a software-independent telemetry representation.
11	GIA	Gateway Integration Architecture	Defines the future adapter boundary between provisioned devices and external gateways or gateway platforms.
12	FRA	Firmware Reproducibility Architecture	Preserves the inputs and outputs required to reproduce a generated firmware build.

2. Full Technical Definitions

1. DPFIA

Device Provisioning & Firmware Integration Architecture

Short definition: Umbrella architecture connecting hardware definition, sensor configuration, firmware generation, provisioning, verification and future gateway integration.

Formal definition: The umbrella architecture defines the separation and interaction of the physical-board, sensor, configuration, firmware, provisioning, verification and integration layers as one device-engineering system.

2. HDCRM

Heterogeneous Device Capability Representation Model

Short definition: Structured representation of supported boards, processors, pins, interfaces, protocols and hardware capabilities.

Formal definition: HDCRM provides a common device abstraction while preserving board-specific characteristics, allowing different microcontroller boards to participate in common configuration, validation and provisioning workflows.

3. SIRM

Sensor Interface Representation Model

Short definition: Structured representation of sensors, measurements, interfaces, wiring requirements, libraries and electrical constraints.

Formal definition: SIRM provides a common sensor abstraction while preserving sensor-specific acquisition characteristics, allowing sensors with digital, analog, I²C, SPI and other interfaces to participate in common configuration workflows.

4. DCRA

Device Configuration Representation Architecture

Short definition: Canonical representation of a complete device configuration, including board, machine/device identity, sensors, parameters, pins, sampling and communication settings.

Formal definition: DCRA defines how a complete physical-device configuration is represented independently from the generated firmware so that configurations can be validated, stored, exported, reproduced and regenerated.

5. APVA

Automated Pin Validation & Assignment Architecture

Short definition: Automatically assigns compatible pins and detects conflicts, reserved pins and interface incompatibilities.

Formal definition: APVA evaluates board capabilities and sensor requirements to produce valid pin assignments, while preventing incompatible allocations and exposing manual overrides when a technician needs a specific assignment.

6. HWVA

Hardware Wiring Validation Architecture

Short definition: Produces deterministic wiring requirements and validates configured connections against board and sensor definitions.

Formal definition: HWVA translates configuration into technician-readable wiring instructions and validates power, ground, signal and interface relationships that are represented by the supported hardware definitions.

7. FGA

Firmware Generation Architecture

Short definition: Transforms validated device configuration into board-specific, sensor-aware firmware source.

Formal definition: FGA separates device configuration from source-code authoring by assembling tested firmware components and configuration values into reproducible firmware projects for supported boards and sensor combinations.

8. FCPA

Firmware Compilation & Provisioning Architecture

Short definition: Compiles generated firmware and provisions it onto the connected physical board using supported toolchains.

Formal definition: FCPA defines the controlled path from generated firmware source through compilation, board/toolchain selection and physical firmware upload, while preserving build and provisioning results for traceability.

9. DVA

Device Verification Architecture

Short definition: Verifies that a provisioned device boots, communicates and produces expected sensor telemetry.

Formal definition: DVA provides post-provisioning verification by observing device startup, communication and sensor outputs and by distinguishing successful provisioning from compilation or upload success alone.

10. LTA

Local Telemetry Abstraction Architecture

Short definition: Normalizes locally observed device readings into a software-independent telemetry representation.

Formal definition: LTA separates physical-device output formats from higher-level application consumers by defining a stable local representation of measurements, timestamps, device identity and parameter values.

11. GIA

Gateway Integration Architecture

Short definition: Defines the future adapter boundary between provisioned devices and external gateways or gateway platforms.

Formal definition: GIA keeps gateway communication independent from core provisioning functions by providing an integration boundary through which gateway configuration, telemetry transport and device registration can later be added without making gateway connectivity a prerequisite for standalone provisioning.

12. FRA

Firmware Reproducibility Architecture

Short definition: Preserves the inputs and outputs required to reproduce a generated firmware build.

Formal definition: FRA connects device configuration, board definition, sensor definitions, firmware template versions and build metadata so that a previously provisioned firmware artifact can be regenerated or audited consistently.

3. Architectural Relationship

- HDCRM defines the board/device capabilities.
- SIRM defines the sensor and measurement capabilities.
- DCRA combines board, sensor, identity, pin, sampling and communication configuration.
- APVA validates and assigns pins using HDCRM and SIRM.
- HWVA converts the validated hardware configuration into deterministic wiring guidance.
- FGA converts the validated configuration into firmware source.
- FCPA compiles and provisions the firmware to the physical board.
- DVA verifies that the provisioned device actually operates and reports expected readings.
- LTA provides a stable local representation of device telemetry.
- GIA is the later boundary for gateway integration and is intentionally not required for standalone provisioning.
- FRA preserves reproducibility and traceability across the configuration/build/provisioning lifecycle.
- DPFIA unifies these constructs as the overall standalone device-engineering architecture.

At the system level, these constructs are unified by DPFIA, the overall Device Provisioning & Firmware Integration Architecture. The intended operational sequence is: device/board definition → sensor definition → configuration → pin validation and assignment → wiring validation → firmware generation → compilation and provisioning → device verification → local telemetry abstraction → optional gateway integration.

4. Standalone-First Boundary

The software is designed to operate independently of any gateway or Maintain AI deployment. A gateway is a later integration target rather than a prerequisite for board detection, configuration, firmware generation, firmware upload or device verification. GIA defines the boundary that permits future gateway adapters without coupling the core provisioning workflow to one gateway implementation.

5. Initial Scope

- Initial boards: Arduino Uno R3 and Arduino Nano, with ESP32 and additional boards added through board definitions.
- Initial sensors: DHT11, SW-420, generic digital input and generic analog input, followed by additional digital, analog, I²C, SPI and other supported interfaces.
- Core workflow: connect → identify/select board → select machine/device context → select sensors/parameters → assign/validate pins → view wiring → generate → compile → upload → verify → save/export configuration.
- Local-first operation: the core provisioning workflow should not require cloud connectivity.
- Gateway integration: added after the standalone provisioning workflow is stable.

6. Naming and Novelty Note

These formal names identify architectural constructs proposed for the software. The names themselves are not claims that the software invented the underlying technologies, protocols, compiler systems, serial communication methods, sensor interfaces or firmware practices. Any academic or public claim of novelty should be supported by implementation evidence, prior-art comparison and experimental validation.

7. Relationship to Maintain AI

The new software is a separate system. It may later provide a gateway integration path into Maintain AI. Maintain AI's existing formal architecture uses MHICPIA as its umbrella architecture and separately defines constructs such as HMIRM, HITSNA, AIITIA and DLE-IIA. This document does not rename or replace those constructs.